Rayan Ahmed

DevOps Engineer · AWS · Kubernetes · GitOps · Platform Engineering · DevSecOps

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PROFESSIONAL SUMMARY

Results-driven Cloud, DevOps & Platform Engineer with **3+ years** of hands-on experience architecting, automating, and operating production-grade AWS infrastructure for SaaS products. Deep expertise in Kubernetes platform engineering (Amazon EKS), Infrastructure as Code (Terraform), GitOps with ArgoCD, end-to-end CI/CD automation using Jenkins, full-stack observability (Grafana, Prometheus, Loki), serverless microservices on ECS Fargate, AI-driven automation, cloud cost optimisation, and production reliability. Proven ability to own complex infrastructure challenges end-to-end, from solution design and implementation to production operations, ensuring high availability, scalability, performance, and adherence to production SLAs.

TECHNICAL SKILLS

Cloud: AWS— EC2, VPC, IAM, RDS, S3, ALB, Auto Scaling, Route53, WAF, CloudFront, ECS, Fargate, Cloud-Watch, Secrets Manager, Backup

Containers & Orchestration: Docker, Kubernetes , Helm, ArgoCD (GitOps)

CI/CD: GitHub Actions, Jenkins, ArgoCD

Infrastructure as Code: Terraform, Ansible

Monitoring & Observability: Grafana, Loki, Prometheus, CloudWatch

Scripting & OS: Python, Bash, Linux (Ubuntu, Amazon Linux)

Databases: MySQL, PostgreSQL, MariaDB, MongoDB, Amazon RDS

Version Control: Git, GitHub, Bitbucket

PROFESSIONAL EXPERIENCE

Senior Cloud/DevOps Engineer · TO THE NEW Pvt Ltd

Noida, India · Jun 2026 – Present

- Architected, owned, and deployed production-grade, microservices-based three-tier applications on **AWS EKS** using **Helm** and **GitOps** with **ArgoCD**. Engineered end-to-end **Jenkins CI/CD** pipelines, implementing rolling updates, zero-downtime deployment strategies, readiness/liveness probes, resource quotas, and Kubernetes RBAC to enable reliable, secure, and scalable application releases.
- Architected and provisioned cloud infrastructure for 10+ SaaS products, designing multi-tier AWS VPC architectures with public/private subnets, NAT Gateways, and IAM RBAC to ensure secure, scalable, and isolated multi-tenant workloads across production environments.
- Strengthened application security by implementing **AWS WAF** managed rule groups (AWSManagedRulesCommonRuleSet, AWSManagedRulesAmazonIpReputationList), geo-blocking, and custom IP whitelisting across product-facing endpoints to mitigate common web exploits and malicious traffic.
- Designed and implemented AWS Backup and disaster recovery strategies for critical production workloads, incorporating automated snapshot policies, cross-region replication, and documented Recovery Time Objective (RTO) and Recovery Point Objective (RPO) targets to improve business continuity and disaster recovery readiness.
- Standardized secrets management across CI/CD pipelines by integrating AWS Secrets Manager, eliminating hardcoded credentials and enforcing least-privilege IAM policies to strengthen security and reduce credential exposure across the platform.
- Owned critical production incidents and escalations across enterprise customer environments, coordinating cross-functional engineering teams to troubleshoot infrastructure and application failures, restore services, drive root cause analysis (RCA), and ensure adherence to Service Level Agreements (SLAs).

- Managed and operated **7 AWS EKS** clusters, establishing end-to-end monitoring and observability using Bitnami Helm charts, and orchestrating Kubernetes cluster upgrades through **Terraform** to ensure platform stability, scalability, and operational consistency across environments.
- Drove the organization's **AI** adoption strategy by defining AI and **Large Language Model (LLM)** initiatives, establishing implementation goals, and coordinating task ownership to accelerate engineering automation and operational transformation.
- Engineered and deployed **AWS Lambda**-based AI automations and LLM-powered solutions across multiple customer environments, streamlining operational workflows, reducing manual intervention, and improving engineering efficiency.
- Led customer-facing technical discussions, presales engagements, solution design conversations, and executive-level escalations, translating business requirements into scalable cloud solutions and driving successful technical delivery.
- Managed and mentored a team of 5 Cloud Engineers, overseeing technical execution, task prioritization, resource allocation, knowledge sharing, and delivery quality across multiple customer engagements.

Cloud/DevOps Engineer · TO THE NEW Pvt Ltd

Noida, India · Feb 2023 – Jun 2026

- Troubleshot, maintained, and upgraded production AWS EKS clusters, ensuring high availability, cluster stability, and zero-downtime operations.
- Achieved approximately 37% reduction in the client's monthly AWS cloud spend, optimizing costs from $60K to $38K through cloud cost analysis and infrastructure optimization.
- Provided 24×7 on-call production support for global clients, managing P1/P2 incidents, performing root cause analysis (RCA), and implementing Corrective and Preventive Actions (CAPA) to improve production reliability.
- Collaborated with development teams to troubleshoot CI/CD deployment failures, debug application and infrastructure issues, and mitigate live production incidents.
- Deployed containerized three-tier applications on AWS EKS using Helm charts, configuring secure ingress, SSL/TLS certificates, and DNS for reliable application access and secure traffic routing.
- Provisioned, configured, managed access to, and maintained MySQL, MariaDB, PostgreSQL, and MongoDB databases, ensuring secure database connectivity, access control, and operational reliability.
- Containerized 10+ applications using Docker and deployed them to Kubernetes, configuring health checks, resource limits, and rolling update strategies to ensure reliable application deployments and efficient resource utilization.
- Managed production AWS infrastructure across EC2, RDS, VPC, IAM, Security Groups, and Application Load Balancers (ALB), maintaining high availability, secure networking, and operational stability across environments.
- Implemented centralized monitoring and logging using Amazon CloudWatch, Prometheus, and Grafana, developing monitoring dashboards and configuring proactive alerting for early incident detection and improved observability.

CLIENT PROJECTS/ AUTOMATIONS

GitOps-Based Kubernetes Platform with ArgoCD

[AWS EKS · ArgoCD · Bitbucket · Helm · Jenkins]

- Designed and implemented an end-to-end GitOps-based CI/CD automation solution for client application deployments on AWS EKS, integrating Jenkins, Bitbucket, Argo CD, Docker, and Amazon ECR to automate application build, image publishing, and Kubernetes deployment workflows.
- Automated Continuous Integration (CI) using Jenkins Declarative Pipelines, Jenkinsfile, and Bitbucket webhooks, triggering automated Docker image builds on code commits and publishing container images to Amazon ECR with image scanning enabled.
- Implemented GitOps-based Continuous Delivery (CD) using Argo CD by integrating Bitbucket repositories as the single source of truth for Kubernetes manifests and Helm charts, enabling automated synchronization of desired and live cluster states and continuous deployment of application and infrastructure configuration changes.
- Configured and deployed Argo CD on AWS EKS using Helm charts, establishing repository connectivity through SSH authentication, Kubernetes application management, automated synchronization, drift detection, self-healing, and Git-based rollback capabilities to improve deployment consistency and operational reliability.

InfraGPT Query Assistant *[AWS Bedrock · Python · EC2 · AWS]*

- Developed InfraGPT, an AI-powered DevOps assistant using Amazon Bedrock (Claude 3 Haiku) that enables developers to query AWS SNS topics, SQS queues, resource configurations, and subscription filter policies using natural language, eliminating manual AWS Console access and reducing dependency on DevOps teams.
- Architected an AWS-native AI infrastructure query solution using Flask (Python), Amazon Bedrock, and Boto3, enabling natural language-to-JSON translation, automated AWS API execution, and conversion of raw infrastructure data into human-readable responses.
- Implemented secure, role-based authentication with Admin and Developer access levels, smart fuzzy matching for mistyped resource names, and command safety validation to improve usability while maintaining controlled access to infrastructure information.
- Designed a cost-efficient and privacy-focused AI automation solution hosted on an existing EC2 instance alongside Grafana, leveraging Amazon Bedrock inference, AWS-managed encryption, TLS-secured API communication, and zero external API calls to keep infrastructure data within the mFine AWS account in the ap-south-1 (Mumbai) region.

AI ALB Logs Analyzer *[AWS Lambda · Python · API Gateway · ALB · Unsupervised Learning]*

- Architected a serverless ETL pipeline utilizing AWS API Gateway and Lambda to stream, decompress, and parse S3 Application Load Balancer logs entirely in-memory with zero disk I/O.
- Engineered a concurrent processing engine using Python ThreadPoolExecutor to parallelize S3 stream ingestion, effectively bypassing API Gateway 29-second synchronous timeout constraints.
- Integrated scikit-learn Isolation Forest unsupervised machine learning algorithms to dynamically baseline time-series traffic buckets and isolate volumetric DDoS IP anomalies.
- Developed a lexical heuristic engine applying Length-Normalized Shannon Entropy and consonant-to-vowel ratios to differentiate malicious directory traversal payloads from legitimate RESTful API endpoints.
- Designed a Layer 7 cache-busting detection module leveraging base path aggregation and query string variance analysis to instantly identify and flag advanced search spam attacks.
- Productized the analytics pipeline into a zero-touch automation triggered via secure cURL REST invocations, instantly generating and dispatching comprehensive HTML diagnostic reports through Amazon SES for rapid incident response

CERTIFICATIONS

AWS Certified Solutions Architect – Associate — Amazon Web Services

AWS Certified Cloud Practitioner — Amazon Web Services, 2024

Microsoft Azure Fundamentals (AZ-900) — Microsoft

Microsoft Data Fundamentals (DP-900) — Microsoft

Microsoft AI Fundamentals (AI-900) — Microsoft, In Progress

AWARDS AND RECOGNITIONS

Star Performer of the Month Award · TO THE NEW · Sept 2023

Impact Award · TO THE NEW · Aug 2025

EDUCATION

Masters of Business Administration (MBA) · Integral University, Lucknow · 2024 – 2026

Bachelor of Technology in Computer Applications (Btech) · Integral University, Lucknow · 2019 – 2023